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RESEARCH INTERESTS	• Early-epoch Galaxy Formation; the Circumgalactic Medium • Spectroscopic Studies of Galaxies and their Environment
EDUCATION	Johns Hopkins University, Baltimore MD, United States Department of Physics and Astronomy, 9/2016 - 12/2022 Ph. D. in Astrophysics; Thesis Advisors: Susan Kassin, Timothy Heckman Thesis title: <i>Observational studies of the relation between galactic winds and star formation at half the age of the Universe</i> Tsinghua University, Beijing, China Department of Physics, 8/2012 - 7/2016 B. Sc. in Physics; Thesis Advisor: Shude Mao
RESEARCH POSITIONS	University of Milano-Bicocca, Milan MI, Italy 12/2022–now Postdoc research fellow (European Research Council funded) Advisor: Sebastiano Cantalupo, Department of Physics Research topics: early galaxy formation in extreme environments & circumgalactic gas University of California, Santa Cruz CA, United States 2020; 7-9/2015 Visiting student; Advisors: Sandra Faber, David Koo, Department of Astronomy Research topics: galactic winds; star formation distribution in moderate-redshift galaxies
PUBLICATION HIGHLIGHTS	• Highlight 1: Early-epoch Galaxy Formation in Extreme Conditions [5] W. Wang, S. Cantalupo, A. Pensabene et al. Nature Astron., 9, 710 (2025): <i>A Giant Disk Galaxy Two Billion Years After The Big Bang</i> (Editorial Highlight) [4] A. Pensabene, S. Cantalupo, W. Wang et al. A&A, 701, A120 (2025): <i>ALMA survey of a massive node of the Cosmic Web at $z \sim 3$: II. A dynamically cold disk galaxy in the proximity of a hyperluminous quasar</i> • Highlight 2: Relations between Galaxies and Circumgalactic Gas [6] W. Wang, S. Cantalupo et al. (<i>to be submitted</i>, 2025): <i>Discovery of a Massive $z > 3$ Quiescent Galaxy at a Gas-Rich Node of the Cosmic Web</i> [3] W. Wang, S. A. Kassin, S. M. Faber, D. C. Koo et al., ApJ, 930, 146 (2022): <i>The Baltimore Oriole's Nest: Cool Winds from the Inner and Outer Parts of a Star-Forming Galaxy at $z = 1.3$</i> • Highlight 3: Early-Epoch Galaxy Star Formation and Dust [2] W. Wang, S. A. Kassin, C. Pacifici et al., ApJ, 869, 161 (2018): <i>Galaxy Inclination and the IRX-β Relation: Effects on UV Star Formation Rate Measurements at Intermediate to High Redshifts</i> [1] W. Wang, S. M. Faber, F.-S. Liu et al., MNRAS, 469, 4063 (2017): <i>UVI colour gradients of $0.4 < z < 1.4$ star-forming main-sequence galaxies in CANDELS: dust extinction and star formation profiles</i> Click this ADS link for full publication list (17 in the past 3 years with 1000+ citations).

AWARDS & MEDIA	Giovani Talenti (Young Talents Awards) by U. Milano-Bicocca & Accademia Nazionale dei Lincei, 2025 Research press releases by the European Space Agency, Nature Portfolio, U. Milano-Bicocca, Caltech, Swinburne U. (reporting W. Wang et al. 2025): “<i>Big Wheel</i>”, a surprisingly large disk galaxy discovered in the early universe. News coverage by Scientific American, The Independent, Newsweek (U.S.), Daily Mail (U.K.), Il Giornale (Italy) (reporting Wang+2025; along with 20+ platforms) The International Astronomical Union Travel Grant, 2019. First-year student award, Johns Hopkins Department of Physics and Astronomy, 2016. National Astronomical Observatory of China Scholarship, 2016. Awards of excellence by the Tsinghua University: 清华学堂计划, 清华领军计划, 2012-2016.
OBSERVATION PROGRAMS (SELECTED)	• Highlight 1: I am heavily involved in a 20-hour JWST NIRSpec program (GO1835) to study $z\sim 3$ galaxies in special environments, with my postdoc advisor as PI. I have been serving as the deputy lead of the planning, implementation, and analysis of observations. These efforts led to two papers led by me and supported five more by other group members. • Highlight 2: I am the core team member to develop an efficient spectroscopic mode for the JWST NIRSpec instrument named Slitlet Stepping. The mode is adopted by two JWST programs of the team (150 hr) and incorporated into the official documentation. • Highlight 3: I am a core member of the VLT MUSE galaxy science team COSMIB, which focuses on studies of the CGM–galaxy connections. Thanks to this, my most recent research was able to be built upon a JWST+MUSE joint approach. ALMA Cycle-12 Program (Co.I.; P.I.: A. Pensabene; 27 hours): <i>The Big Wheel Galaxy: a unique laboratory to study the physics of early disk formation in overdense environments</i>, scheduled for 2025-2026 JWST Cycle-4 Program (Co.I.; P.I.: B. Wang; 20 hours): <i>Extremely massive galaxies in the early universe? Confirming the nature of the most model-breaking object by hunting for stellar absorption features</i>, scheduled for 2025-2026 VLT P112 Program (Co.I. & core team; P.I.: Cantalupo; 84 hours): <i>Connecting the dots with MUSE: the Cosmic Web in emission around a massive structure at $z=3$</i>, 2024-2025 JWST Cycle-1 Program (Co.I. with major contributions, core team; P.I.: Kassin; 74 hours): <i>A Pathfinder for JWST Spectroscopy: Deep High Spectral Resolution Maps of Galaxies over $1 < z < 6$</i>, 2023 JWST Cycle-1 Program (joined in 2022 with major contributions, core team; P.I.: Cantalupo; 24 hours): <i>Unraveling the Knots of Gaseous Cosmic Web Filaments at $z\sim 3$ through H-alpha Emission Observations</i>, 2023 ALMA Cycle-7 Program (P.I.) (15 hours): <i>Does molecular gas follow the motion of ionized gas inside typical high-redshift star-forming galaxies?</i> Observations not completed due to weather and the impact of COVID-19 in Chile, 2021 NASA ADAP Proposal (Co.I. with major contribution; P.I.: Kassin; \$485k): <i>Expelling Gas from Galaxies in the Distant Universe</i>, 2020-2022 Summed metrics: 180+ hours on JWST as core or founding team member; 100+ hours on the Very Large Telescope (VLT) and Hubble Space Telescope (HST) as core team member.
COLLABORATIONS & SERVICES	Research Collaborations: • COSMIB (core team; VLT/JWST), • GARDEN/MEADOW (founding & core team; JWST), • SPRING (JWST), • CEERS (JWST), • HALO7D (core team until 2022; Keck). SOC/LOC member, Conference What Matter(s) Around Galaxies, 2024

CONFERENCES AND TALKS (SELECTED)	The Baryon Cycle from Reionization to the Cosmic Noon, Chile, 2025 Tracing Cosmic Evolution with Galaxy Clusters, Italy, 2025 <i>(Invited talk)</i> From Fake News to Real Clusters: The Controversial Fate of High-z Galaxy Protoclusters, Spain, 2025 Observing and Simulating Galaxy Evolution in the Era of JWST, Switzerland, 2024 <i>(Invited talk)</i> Galaxies and Diffuse Gas in Large-Scale Overdense Environments, Italy, 2024 <i>(SOC/LOC member)</i> What Matter(s) Around Galaxies 2024, Lake Como, Italy, 2024 Seminars, Tsinghua University/Peking University/NAOC, Beijing, China, 2023/2024 Astronomy Seminar, University of California, Riverside, U.S. (remote), 2021 Steward/NOIRLab Galaxy Group Lunch Talk, University of Arizona, U.S. (remote), 2021 Massively Parallel Large Area Spectroscopy from Space, Portugal (remote), 2021 Astrophysics Seminar at University of Missouri, U.S. (remote), 2020 Santa Cruz Galaxy workshop, Santa Cruz, California U.S., 2018
MENTORSHIP	Ying Qin, Johns Hopkins undergraduate in physics major, 2021-2023 (thesis project) <i>Studying the Mg II emission and leaking ionizing photons from low-mass galaxies at $z \sim 1$. Resulting in two conference poster presentations, including one at the AAS Annual Meeting; Now pursuing a career as lawyer in Washington D.C.</i> M. Francesca Uboldi, Bicocca undergraduate in physics major, 2024 (thesis project): <i>Relations between galaxy colors and morphology based on the JWST medium-band data. Successful bachelor's degree defense; Now pursuing Masters in Astronomy degree in Bicocca.</i> Xiaohan Wang, Visiting Ph.D student of Tsinghua University, 2025: <i>JWST NIRSpec studies of galaxies in the high-redshift cosmic web. Relations between galaxy colors and morphology based on the JWST medium-band data. Resulting in one research paper completed & to be submitted to A&A.</i>
TEACHING EXPERIENCE	Laboratory of Data Analysis for Master Students in Astrophysics University of Milano-Bicocca, Spring 2023 Teaching Assistant, General Physics I for Biological Science Majors (171.103) Johns Hopkins University, Fall 2016 Teaching Assistant, General Physics Laboratory (171.111) Johns Hopkins University, Fall 2016
OUTREACH ACTIVITIES	Science outreach day at University of Milano-Bicocca, Italy 2024 <i>A community event open to students from local primary/middle schools in Milan; co-designing the exhibition about astrophysical science with JWST</i> Member of the Astro Scholars program 2021-2022 <i>An annual week-long program about astrophysics and computer programming for undergraduates from under-represented backgrounds; serving as a core member of the hiring & education team; monthly tag-up with the students during the rest of the year</i> Member of the Physics and Astronomy Graduate Students (PAGS) Outreach Team, Johns Hopkins University, 2017-2019 <i>Supporting visits of students from Baltimore local primary/middle schools around once per semester and teaching fundamental physics with educational demos</i> Volunteer teacher at the Pengzhai Primary School, Guizhou, China, Summer/2013 <i>Teaching multiple STEM-related courses for Grade 3-6; the school, with very limited resources, is located in one of the least developed areas of the country.</i>